# Ze-Yu Xing

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Beijing, China
Website · GitHub
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PhD Candidate in Physics

## RESEARCH PROFILE

PhD researcher working primarily on open quantum systems and non-equilibrium quantum dynamics. Research interests include quantum many-body dynamics, relaxation and transport, dissipation and decoherence, non-Hermitian physics, statistical mechanics, quantum information, quantum simulation, and connections with machine learning.

## EDUCATION AND AFFILIATION

Aug. 2022 - Present
**PhD Candidate in Physics**
Institute of Physics, Chinese Academy of Sciences · Beijing, China

Aug. 2018 - Jun. 2022
**B.Sc. in Physics, Honors Program in Science**
Jilin University · Jilin, China

GPA: 3.88/4.00 (93.2/100).

## RESEARCH INTERESTS

- Non-equilibrium quantum dynamics
- Quantum many-body physics
- Statistical mechanics
- Open quantum systems
- Non-Hermitian physics
- Quantum information and simulation

## ONGOING RESEARCH PROJECTS

### OYA-OQS: Liouvillian Lieb-Schultz-Mattis Constraints in Open Quantum Systems ↗

Investigating spectral constraints in translationally invariant open quantum many-body systems with strong U(1) symmetry. The project combines analytical Weyl-sequence arguments with momentum-resolved exact diagonalization and tensor-network simulations of dissipative spin chains.

### ssb-qme: Symmetry-Enforced Strong Quantum Mpemba Effect ↗

Developing a symmetry-based mechanism for strong quantum Mpemba effects in symmetry-broken open systems. Current work studies discrete parity and continuous U(1) symmetries through Liouvillian analysis and quantum-master-equation simulations.

## PUBLICATIONS AND PREPRINTS

2026 **Geometry-adaptive formulation of non-Bloch bands in arbitrary dimensions and spectral instability** ↗
**Ze-Yu Xing**, Yuncheng Xiong, and Haiping Hu

*Communications Physics 9, 127 (2026)* · arXiv:2407.01296 (2024)
Ze-Yu Xing and Yuncheng Xiong contributed equally.

2025 **Universal spreading dynamics in quasiperiodic non-Hermitian systems** ↗
**Ze-Yu Xing**, Shu Chen, and Haiping Hu

*Physical Review B 111, L180203 (2025)*

## HONORS AND AWARDS

2020 **Ny Tsi-Ze Scholarship**
Institute of Physics, Chinese Academy of Sciences

2020 **Honorable Mention, Mathematical Contest in Modeling**
Consortium for Mathematics and Its Applications

2019 **National Scholarship**
Ministry of Education, People's Republic of China

2019 **Outstanding Student of Jilin University**
Jilin University